A Level Maths > Questions >

Functions and Inverse Functions

A-level Maths Topic Summaries - Functions

Subject & topics: Maths | Functions | General Functions **Stage & difficulty:** A Level P2

Fill in the blanks to complete the summary notes on functions and inverse functions.

Part A

Mappings and functions

A **mapping** turns members of an input set into members of an output set. **Figure 1** illustrates four different types of mapping.

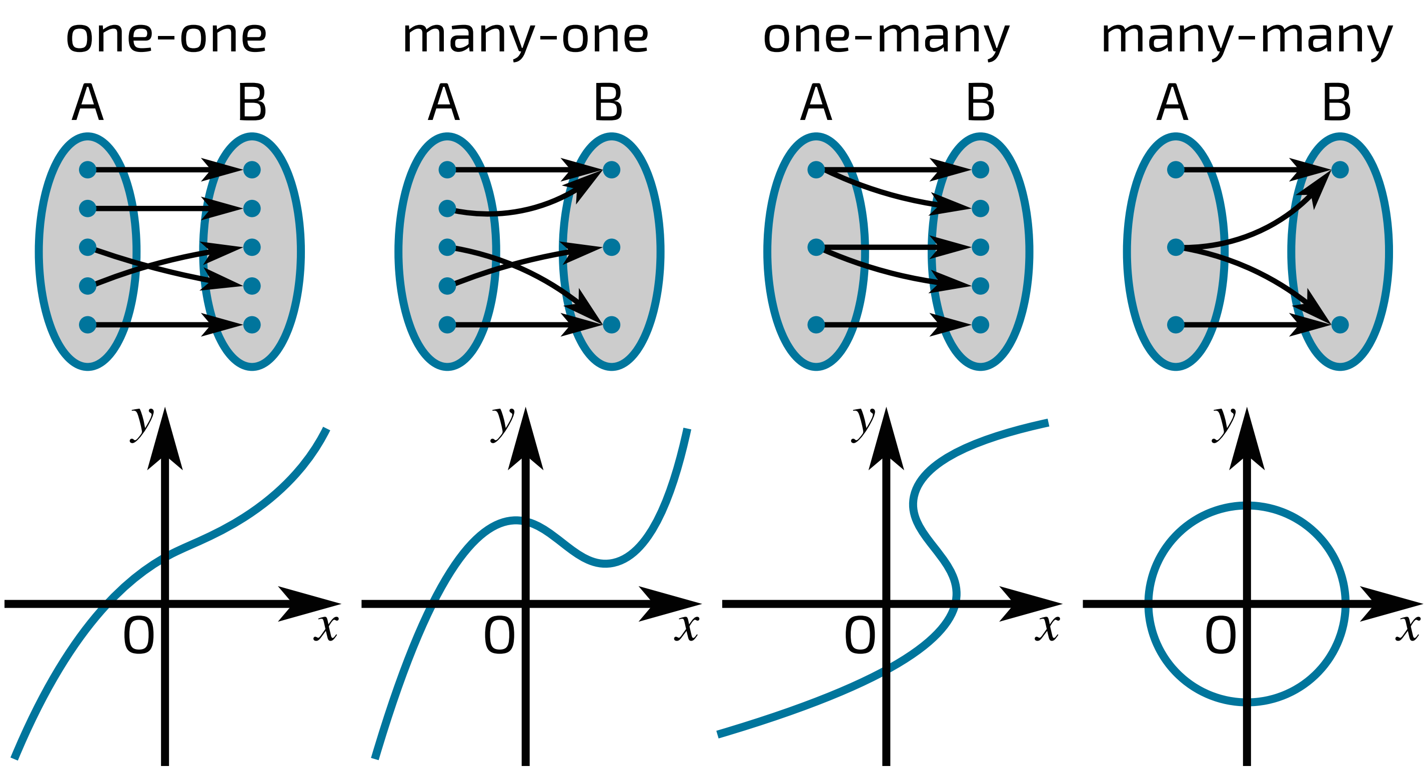

Figure 1: Illustrating different types of mapping. For each type there is an abstract illustration (input set A to output set B), and a graphical illustration (input x to output y).

Complete the table with the correct types of mapping.

Inputs	Mapping
Each input maps to one output. Each output is mapped to by at most one input.	
Each input maps to one output. There are outputs mapped to by more than one input.	many-one
There are inputs that map to more than one output. Each output is mapped to by at most one input.	
There are inputs that map to more than one output. There are outputs mapped to by more than one input.	

Functions are mappings for which each input is mapped to one output. One-one and ____ mappings are functions.

Items:

many-many | many-one | one-many | one-one

Part B

Describing and combining functions

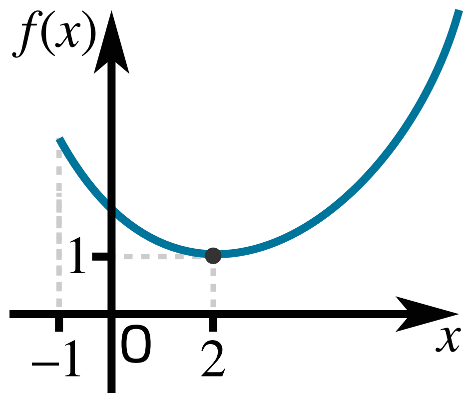

Figure 2: A sketch of the function $f(x) = \frac{1}{4}x^2 - x + 2$, $x \geq -1$.

When we define a function, the **domain** is the set of values that can be put [ ] the function. The **range** is the set of possible [ ] values of the function. For example, let us define the function $f(x)$ as

$$f(x) = \frac{1}{4}x^2 - x + 2, \qquad x \in \mathbb{R}, \;\; x \geq -1$$

This function is shown in **Figure 2**. The domain of $f(x)$ is stated in the definition: [ ] (x is a real number which is greater than or equal to -1). The range is [ ] (the output of the function is greater than or equal to 1).

Composition of functions is where the output of one function is used as the input to a second function. For example, if the output of a function $g(x)$ is used as the input of another function $h(x)$, we could write this as [ ]. We could also write this as $hg(x)$ or $h \circ g(x)$, where the function applied first is the function furthest to the [ ]. In order for a composite function to exist, the [ ] of the first function must be within the [ ] of the second function.

It is also possible to compose a function with itself. We would write this as $g\big(g(x)\big)$ or [ ]. This is **not** the same as squaring a function, $\big(g(x)\big)^2$. The notation can be ambiguous, so if in doubt use brackets to make the meaning clear.

Items:

domain | into | left | output | range | right | $f(x) \geq 1$ | $x \in \mathbb{R}, \;\; x \geq -1$ | $h\big(g(x)\big)$ | $g^2(x)$

Part C

Inverse functions

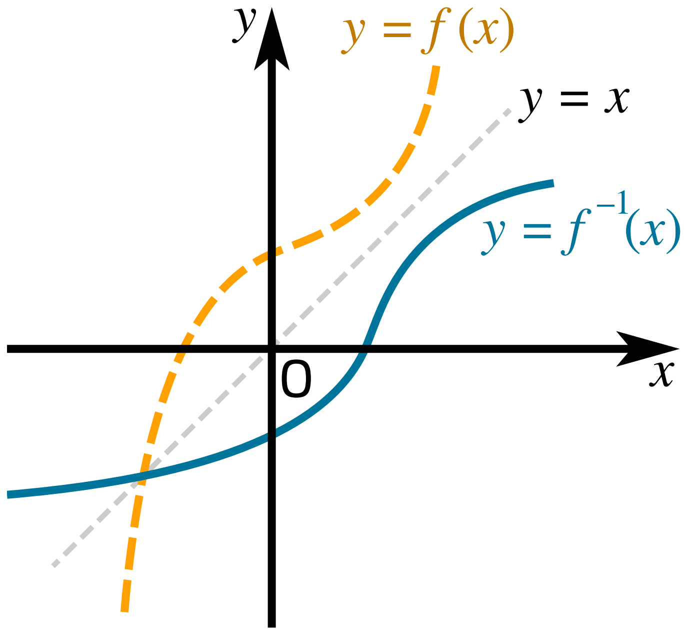

Figure 3: A function $f(x)$ (yellow dashed line) and its inverse function $f^{-1}(x)$ (blue solid line).

For [ ] functions, each input produces a unique output. This means that if we are given a value for the output of the function, we can work out exactly what the input to the function was. We can define a new function which "reverses" the original function. We call this an **inverse function**, and we write it as $f^{-1}(x)$. Note that $f^{-1}(x)$ does **not** mean the reciprocal of the function, $\left(f(x)\right)^{-1} = \frac{1}{f(x)}$.

The composition of a function and its inverse returns the value that we started with,

$$f\left(f^{-1}(x)\right) = f^{-1}\left(f(x)\right) = \boxed{}$$

In effect, in going from a function to its inverse, we swap inputs and outputs. The output of the function, $f(x)$, becomes the input to the inverse function, x. Similarly, the input to the function, x, becomes the output of the inverse function, $f^{-1}(x)$. This has several consequences:

The domain of $f^{-1}(x)$ is the [ ] of $f(x)$.
The range of $f^{-1}(x)$ is the [ ] of $f(x)$.
Visually the graph of $f^{-1}(x)$ is the [ ] of the graph of $f(x)$ in the line [ ]. Figure 3 shows an example.

Items:

domain | one-one | range | reflection | x | $y = x$

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Functions and Algebra 5ii

Subject & topics: Maths **Stage & difficulty:** A Level C1

The function f is defined by

$$f(x) = \frac{1}{\sqrt{x}} + 2, \quad x > 0.$$

The function g is defined for all real values of x by

$$g(x) = 10 - (x+3)^2.$$

Part A

Range of f

State the range of $f(x)$ as an inequality.

The following symbols may be useful: `<, <=, >, >=, f(x), x, y`

Part B

Inverse of f

Find an expression for $f^{-1}(x)$.

The following symbols may be useful: `f, x`

Part C

Range of g

State the range of $g(x)$ as an inequality.

The following symbols may be useful: `<`, `<=`, `>`, `>=`, `g(x)`, `x`, `y`

Part D

Compound function of g

Find the value of $g\big(g(-1)\big)$.

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Inverse Quadratic Function

Subject & topics: Maths | Functions | Graph Sketching **Stage & difficulty:** A Level P2

Figure 1 shows the graph of $y = f(x)$, where

$$f(x) = 2 - x^2, \quad x \leq 0$$

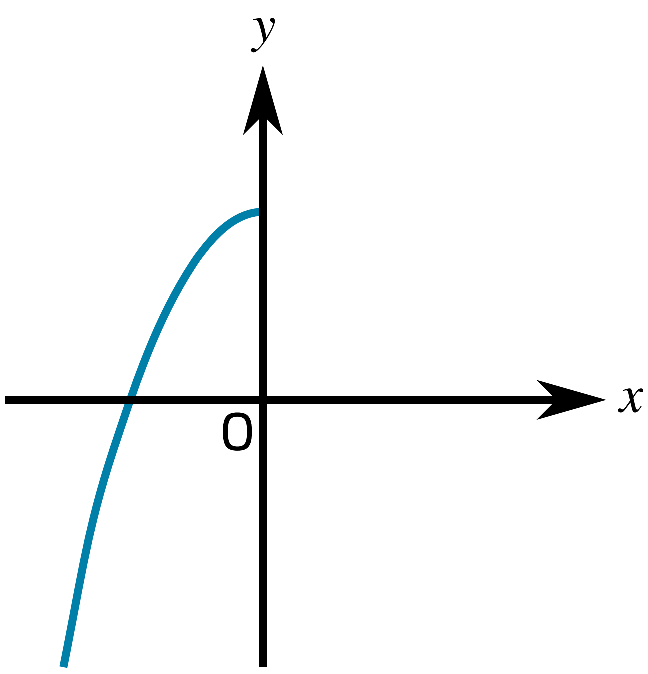


Figure 1: The graph of $y = f(x)$, for $x \leq 0$.

Part A

$f^2(-3)$

Evaluate $f^2(-3)$.

Part B

$f^{-1}(x)$

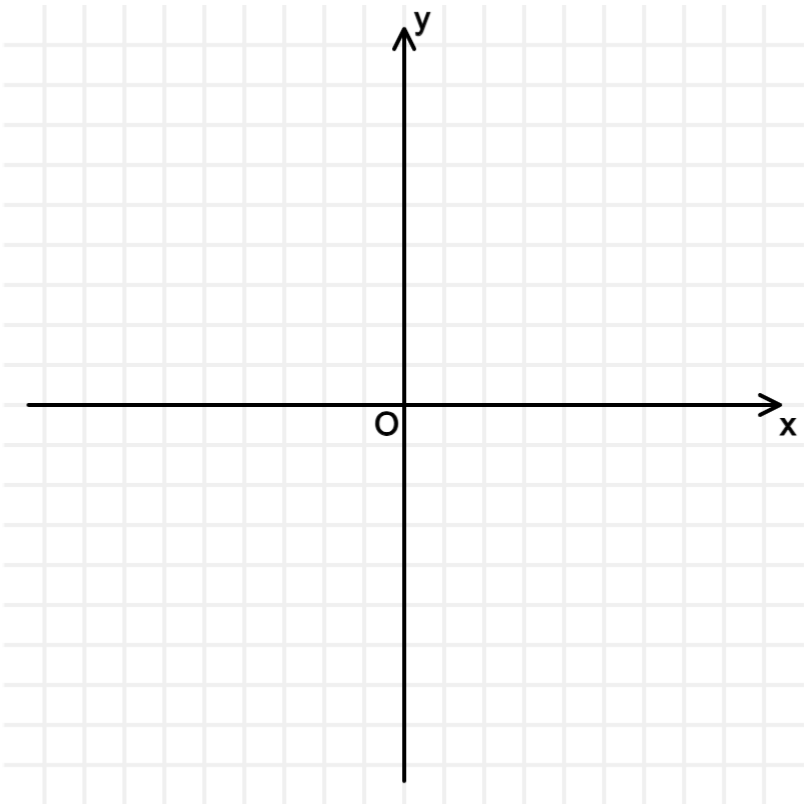
Find an expression for $f^{-1}(x)$.

The following symbols may be useful: `f, x, y`

Part C

Graph of $f^{-1}(x)$

Sketch the graph of $y = f^{-1}(x)$.



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The Modulus Function

A-level Maths Topic Summaries - Functions

Subject & topics: Maths | Functions | General Functions **Stage & difficulty:** A Level P2

Fill in the blanks to complete the summary notes on the modulus function.

Part A

The modulus function, $f(x) = |x|$

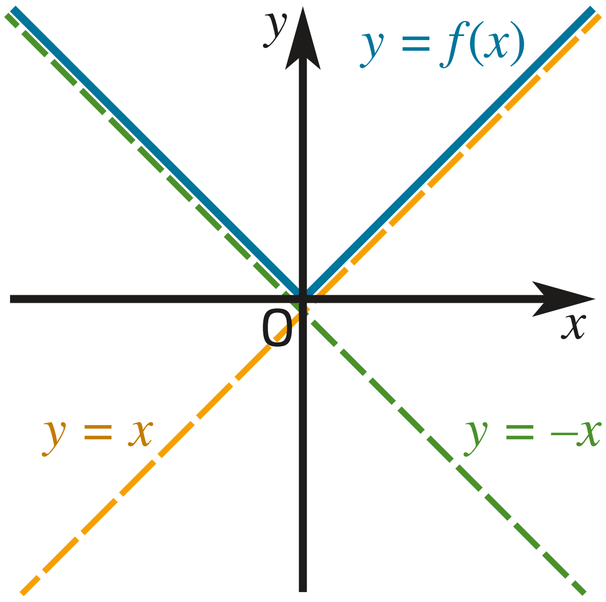

Figure 1: Illustrating the modulus function. Solid blue line: $y = |x|$. Dashed yellow line: $y = x$. Dashed green line: $y = -x$.

The modulus function $|x|$ is defined as

$$|x| = \begin{cases} \boxed{} & x \geq 0 \\ \boxed{} & x < 0 \end{cases}$$

The modulus function leaves positive values of x [ ]. The modulus function [ ] of negative values of x. Here are some numerical examples:

$$\begin{aligned} |3| &= \boxed{} \\ |-7| &= \boxed{} \\ |0| &= \boxed{} \end{aligned}$$

Figure 1 shows a graph of the modulus function. For positive values of x, $y = |x|$ is the same as [ ]. For negative values of x, $y = |x|$ is the same as [ ].

Items:

0 | 3 | 7 | $-x$ | x | $y = -x$ | $y = x$ | changes the sign | unchanged

Part B

Applications: $|g(x)|$ and $g(|x|)$

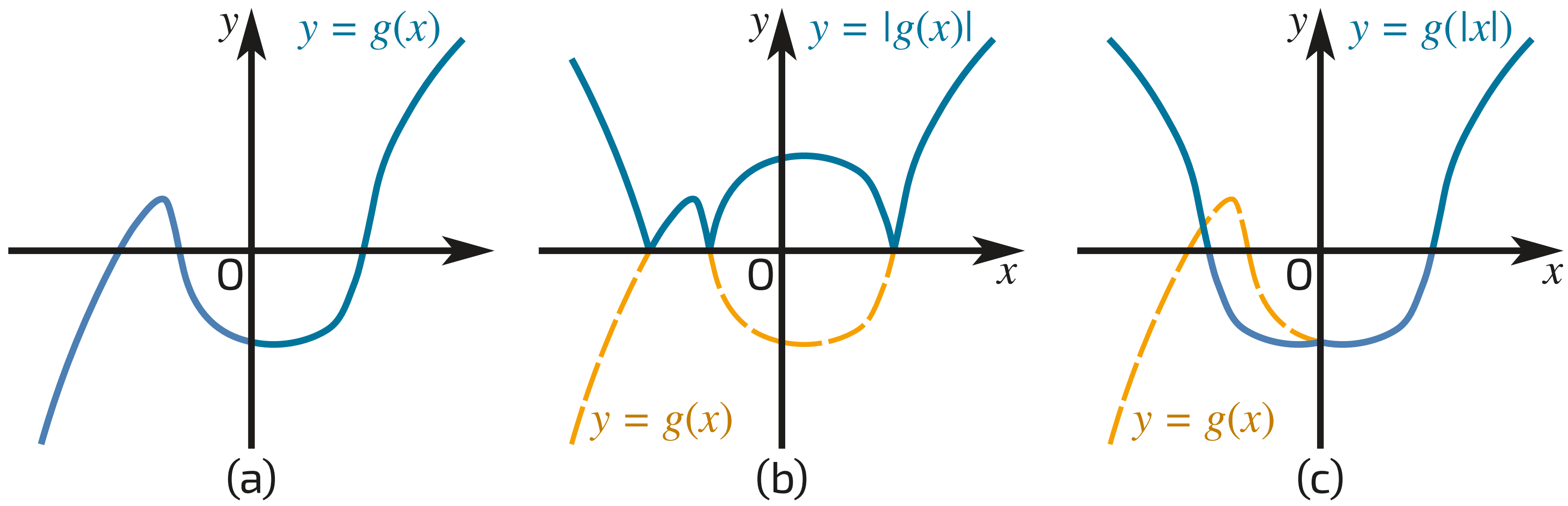

Figure 2: The graphs of (a) $y = g(x)$, (b) $y = |g(x)|$ and (c) $y = g(|x|)$.

Case 1: $|g(x)|$

When modulus signs are applied to the output of a function, the modulus signs are applied after the value of the function is calculated. The effect is that parts of $g(x)$ above the x-axis are left [ ], while parts of $g(x)$ below the x-axis are [ ] in the [ ]-axis.

Case 2: $g(|x|)$

When modulus signs are applied to the input of a function, the modulus signs are applied [ ] the value of the function is calculated. This means that the output of the function when $x = -a$ will be the same as the output of the function when $x =$ [ ]:

$$g(|-a|) = g(a)$$

The effect is that the part of $g(x)$ for $x < 0$ is a [ ] of the part of $g(x)$ for $x > 0$ in the [ ]-axis.

Items:

0 a x y before reflected reflection unchanged

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Function Types and Inverses

Subject & topics: Maths | Functions | General Functions **Stage & difficulty:** A Level C1

Figure 1 shows five different graphs, A, B, C, D and E, each for values of x such that $-a \leq x \leq a$ where a is a constant.

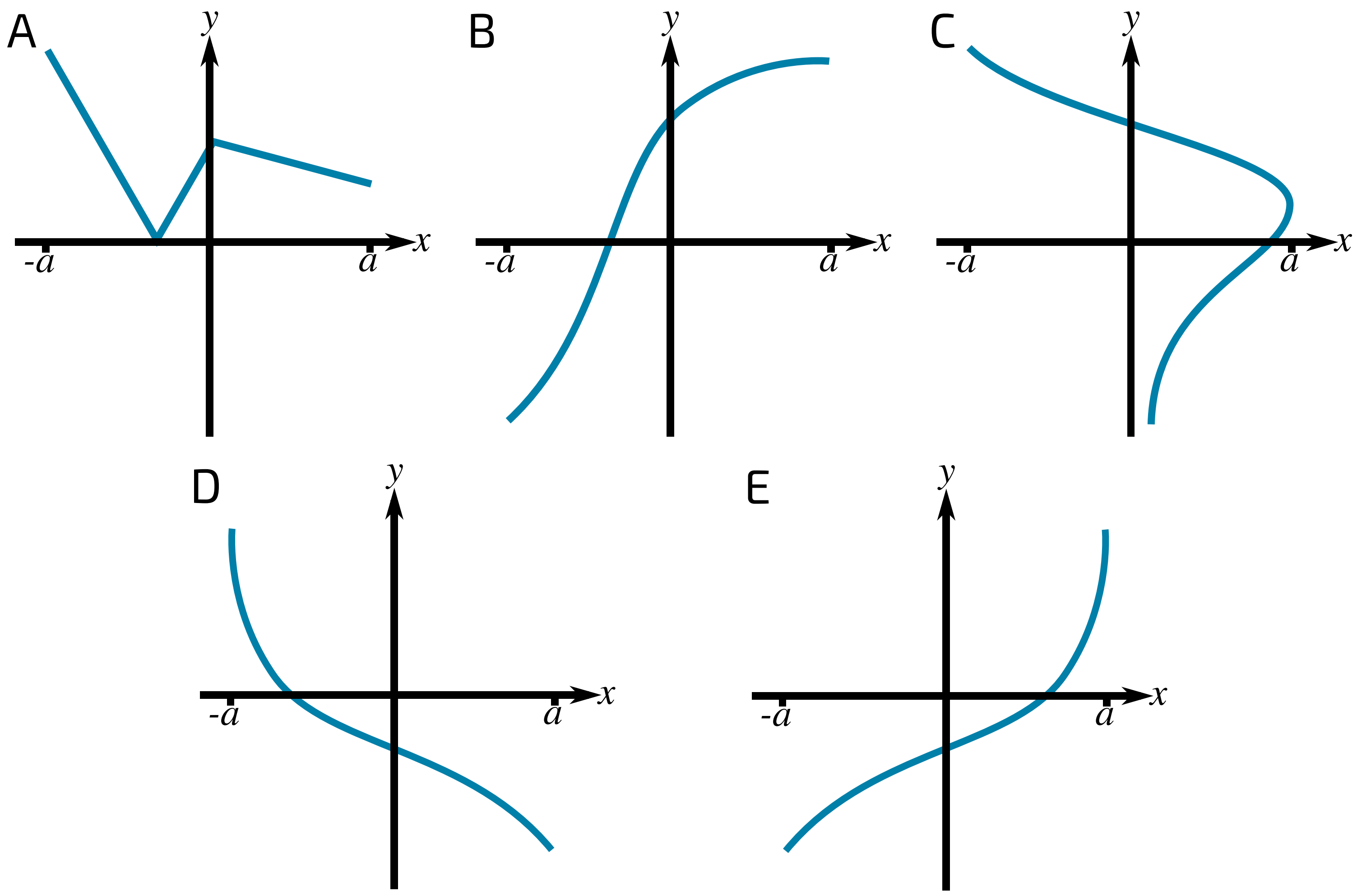


Figure 1: The set of five graphs, labelled A, B, C, D and E

Part A

Function

Which diagram does not show the graph of a function?

- A
- B
- C
- D
- E

Part B

One-to-one Function

Which diagram shows the graph of a function that is not one-to-one?

- A
- B
- C
- D
- E

Part C

Inverses

It is given that two of the diagrams illustrate functions that are inverses of each other. Identify one of these two diagrams.

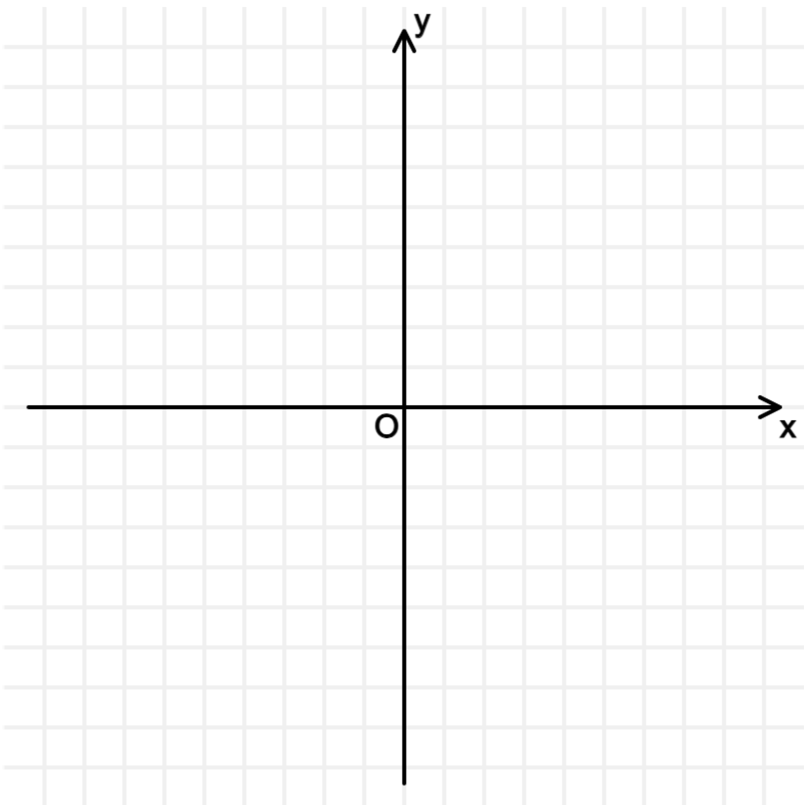
- A
- B
- C
- D
- E

Part D

Sketch

The graph in E has equation $y = f(x)$. Sketch the graph of $y = \left|f(x)\right|$.

To prevent any sharp changes in your curve from being smoothed out, sketch your curve as two sections.



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Functions and Algebra 3i

Subject & topics: Maths **Stage & difficulty:** A Level P2

The functions f and g are defined for all real values of x by

$$f(x) = |2x + a| + 3a \quad \text{and} \quad g(x) = 5x - 4a,$$

where a is a positive constant.

Part A
Range

Find the range of $f(x)$.

Fill in the inequality below.

Items:

$f(x)$ $<$ $\leq$ $>$ $\geq$ $< f(x) <$ $\leq f(x) \leq$ $< f(x)$ or $f(x) <$ $\leq f(x)$ or $f(x) \leq$ $\frac{a}{3}$ $\frac{a}{2}$ a $2a$ $3a$ $4a$ 0 $-\frac{a}{3}$ $-\frac{a}{2}$ $-a$ $-2a$

Part B

Inverse function of $f(x)$

Fill in the blanks to explain why the function $f(x)$ has no inverse.

The function $f(x)$ is not [ ]. For example, $f(0) = 4a$ and $f($[ ]$)$ also equals $4a$. Hence, $f(x)$ has no inverse.

Items:

many-to-one | one-to-one | $2a$ | one-to-many | a | many-to-many | $-2a$ | $-a$

Part C

Inverse function of $g(x)$

Find an expression for $g^{-1}(x)$.

The following symbols may be useful: a, x

Part D

Solve for x

Solve for x the equation $g\big(f(x)\big) = 31a$. Give your solutions in ascending order.

$x =$ [ ] a

$x =$ [ ] a

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Modulus 3ii

Subject & topics: Maths **Stage & difficulty:** A Level P1

Solve the inequality $|2x-5| > |x+1|$.

Construct your answer from the items below.

Items:

x $<$ $\leq$ $>$ $\geq$ $< x <$ $\leq x \leq$ $> x$ or $x >$ $\geq x$ or $x \geq$ -6 -4 -2 $-\frac{3}{2}$ $-\frac{4}{3}$ -1 $-\frac{1}{2}$ 0 $\frac{1}{2}$ 1 $\frac{4}{3}$ $\frac{3}{2}$ 2 4 6

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Modulus Functions 2

Pre-Uni Maths for Science E4.6

Subject & topics: Maths | Functions | General Functions **Stage & difficulty:** A Level P2

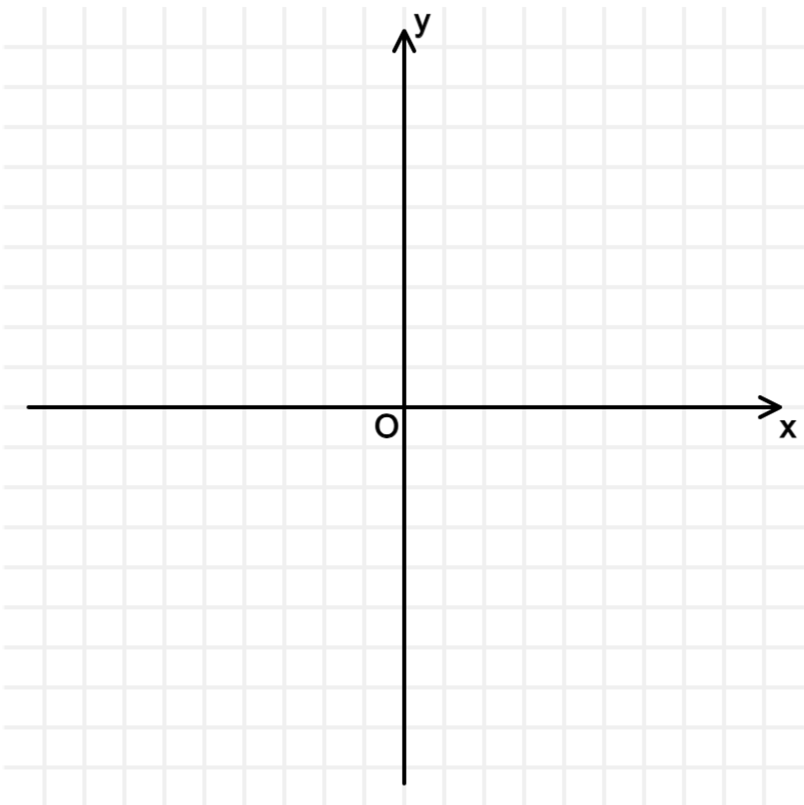
Part A

$y = |\frac{1}{x}|$

State the value of x where the function $y = \left|\frac{1}{x}\right|$ diverges.

Sketch the graph of $y = \left|\frac{1}{x}\right|$.

You do not need to draw the asymptotes.



Part B

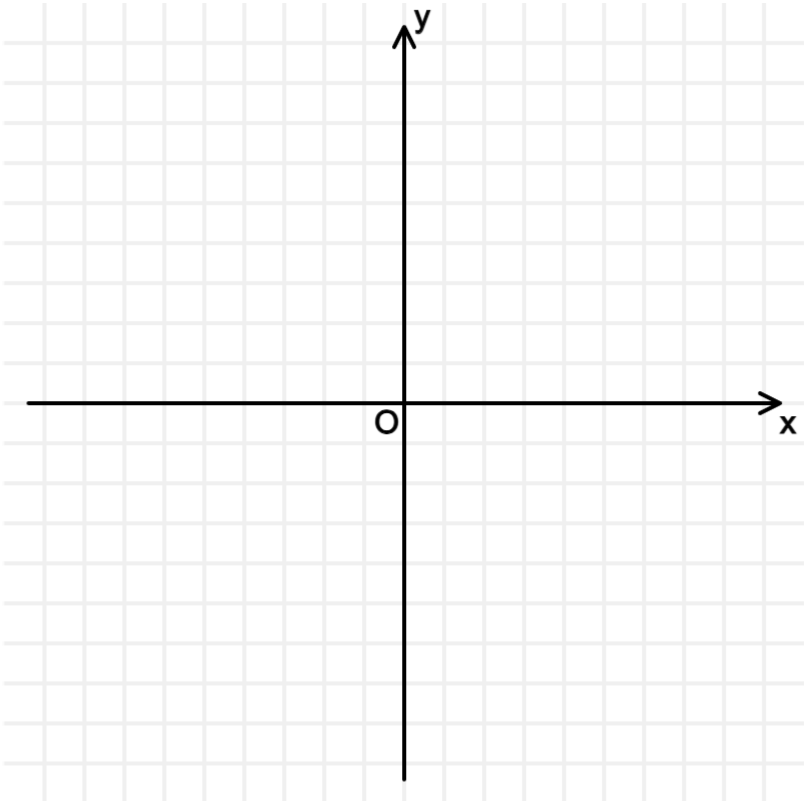
$y = \left|\frac{1}{x^2-4}\right|$

State the values of x where the function $y = \left|\dfrac{1}{x^2-4}\right|$ diverges.

The following symbols may be useful: x, ±

Sketch the graph of $y = \left|\dfrac{1}{x^2-4}\right|$.

You do not need to draw the asymptotes.



Part C

Solve equation graphically

Solve the equation $|x| = \left|\dfrac{1}{x}\right|$ graphically and give the solution as a single expression.

The following symbols may be useful: x, ±

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Curve Sketching and Combined Transformations 3i

Subject & topics: Maths **Stage & difficulty:** A Level P2

The function f is defined for all real values of x by

$$f(x) = k(x^2 + 4x)$$

where k is a positive constant. **Figure 1** shows the curve with equation $y = f(x)$.

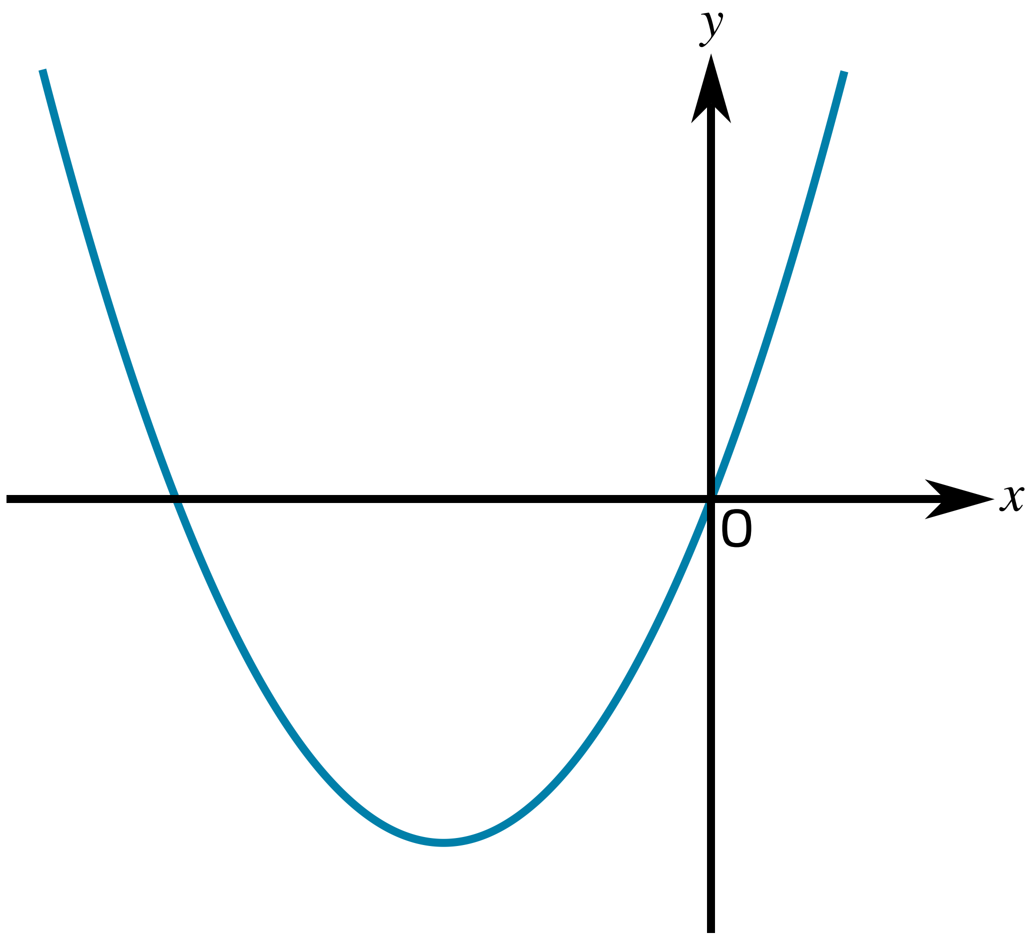

Figure 1: The graph of $y = f(x)$

Part A

Transformations

Give a sequence of transformations that will transform the curve $y = x^2$ to the curve $y = f(x)$.

Available items

- Translate the curve 4 units in the positive x direction.
- Translate the curve 2 units in the positive x direction.
- Stretch the curve in the x direction by a factor of $\frac{1}{k}$.
- Translate the curve 4 units in the negative x direction.
- Stretch the curve in the y direction by a factor of k.
- Stretch the curve in the x direction by a factor of k.
- Translate the curve 4 units in the positive y direction.
- Translate the curve 4 units in the negative y direction.
- Translate the curve 2 units in the negative x direction.
- Translate the curve 2 units in the positive y direction.
- Translate the curve 2 units in the negative y direction.
- Stretch the curve in the y direction by a factor of $\frac{1}{k}$.

Part B

Range

Find the range of $f(x)$ as a single inequality in terms of k.

The following symbols may be useful: `<, <=, >, >=, f(x), k, x, y`

Part C

Find k

It is given that there are three distinct values of x which satisfy the equation $\left|f(x)\right| = 20$.

Find the value of k.

Part D

Solve $\left|f(x)\right| = 20$

Using the value of k from part C, find the three distinct values of x which satisfy the equation $\left|f(x)\right| = 20$. Give any irrational values to 3 sf.

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Combined Transformations

Subject & topics: Maths | Functions | General Functions **Stage & difficulty:** A Level P2

The function f is defined by $f(x) = \sqrt{mx+7} - 4$, where $x \geqslant -\frac{7}{m}$ and m is a positive constant. **Figure 1** shows the curve $y = f(x)$.

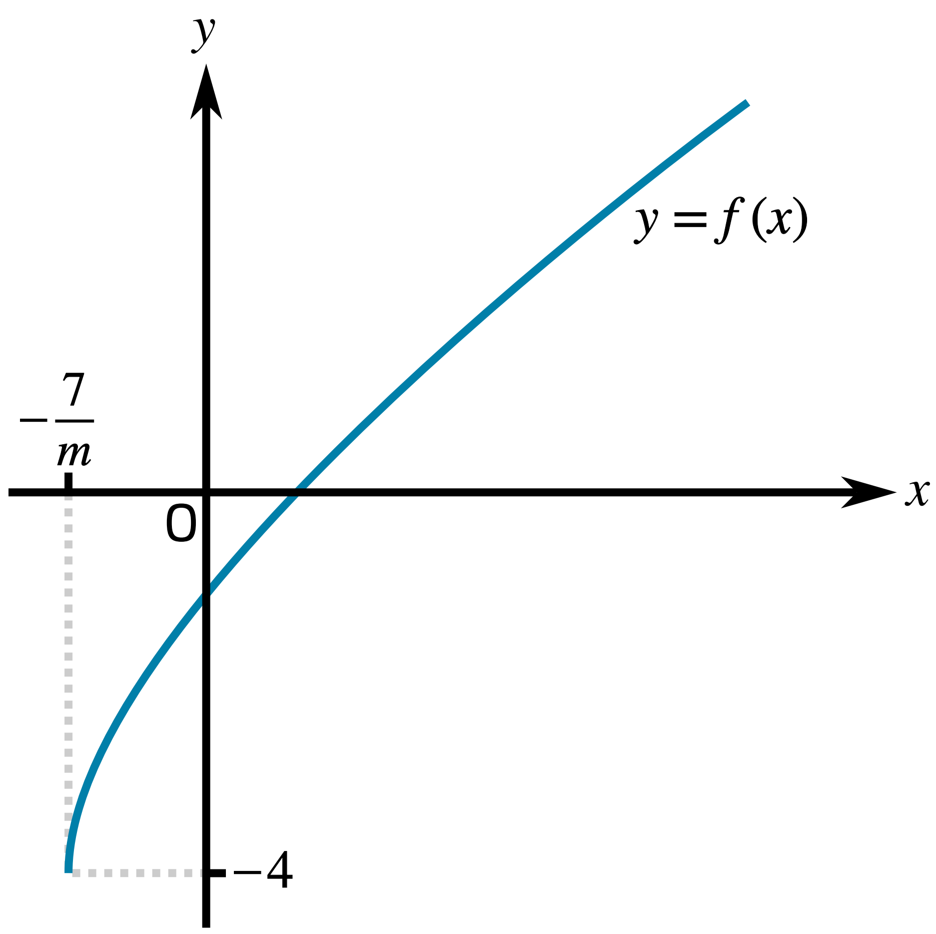

Figure 1: The curve $y = f(x)$

Part A

Translation of the curve $y = \sqrt{x}$

A sequence of transformations maps the curve $y = \sqrt{x}$ to the curve $y = f(x)$. Give details of these transformations.

Available items

- Stretch the curve in the y direction by a factor of $\frac{1}{m}$.
- Translate the curve 7 units in the positive x direction.
- Translate the curve 7 units in the negative y direction.
- Stretch the curve in the x direction by a factor of $\frac{1}{m}$.
- Translate the curve 4 units in the positive y direction.
- Translate the curve 4 units in the negative x direction.
- Translate the curve 4 units in the negative y direction.
- Stretch the curve in the x direction by a factor of m.
- Translate the curve 7 units in the negative x direction.

Part B

$f^{-1}(x)$

Find an expression for $f^{-1}(x)$.

The following symbols may be useful: `f`, `m`, `x`

Part C

Values of m

It is given that the curves $y = f(x)$ and $y = f^{-1}(x)$ do not meet. Thus it can be deduced that neither curve meets the line $y = x$. Hence determine the possible values of m.

Construct your answer from the items below.

Items:

m $<$ $\leq$ $>$ $\geq$ $< m <$ $\leq m \leq$ $> m$ or $m >$ $\geq m$ or $m \geq$ -28 -14 -8 -7

-4 -2 -1 0 1 2 4 7 8 14 28

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